

Tony Li

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EDUCATION

- 08/2024–Present **Stony Brook University**
Ph.D. in Computer Science; focus: Human-AI Interaction, LLM and GenAI
- 08/2020–06/2024 **University of Science and Technology of China (USTC)**
B.E. in Computer Science and Technology

PUBLICATIONS

- Under Review **Li, Tony**, CHEN, Z., ZHU, P., MA, Y., AND BI, X. LEXGR: Extending Flexible LLM-Based Keyboard Decoding to New Words
- Under Review CHEN, Z. *, **Li, Tony** *, ZHU, P., AND BI, X. DecoLego: A Functional Framework for Probing, Recomposing, and Extending Keyboard Decoders
- Under Review NEWTON, F., **Li, Tony**, BI, X., AND LEIVA, L. A. Decoding without Profiling: Privacy-Preserving Gesture Typing through Backtracking
- Under Review CHEN, Z., ZHU, P., CHEN, Y., **Li, Tony**, GU, X. D., MIAO, X., YU, L., AND BI, X. Structure Before Prose: A Persistent Argument Representation for Source-Grounded Legal Writing
- CHI'26 **Li, Tony**, MA, Y., LI, Z., YU, C., RAMAKRISHNAN, I. V., AND BI, X. KeySense: LLM-Powered Hands-Down, Ten-Finger Typing on Commodity Touchscreens
- CHI'25 XU, W., **Li, Tony**, WANG, Y., YANG, X.-D., AND WU, T.-Y. BIT: Battery-free, IC-less and Wireless Smart Textile Interface and Sensing System
- VRW'25 MA, Y., **Li, Tony**, LI, Z., AND BI, X. LLM-powered Text Entry in Virtual Reality
- CHI EA'25 XU, W., **Li, Tony**, WANG, Y., YANG, X.-D., AND WU, T.-Y. Demonstrating BIT: Battery-free, IC-less and Wireless Smart Textile Interface and Sensing System

EXPERIENCE

- 06/2026–08/2026 **GenAI Research Intern, Samsung Ads**, Mountain View, CA
- Developed reusable **user embedding** methods for large-scale advertising systems, targeting efficient personalization under real-world serving constraints.
 - Explored **LLM/GenAI-based user representation learning**, knowledge distillation, and agentic personalization for next-generation Ads Intelligence systems.
- 08/2024–Present **Graduate Research Assistant, Stony Brook University**, Advisor: Xiaojun Bi
- Developed **LLM-based keyboard decoders** supporting personalized vocabulary and co-led a framework for analyzing and extending decoder capabilities.
 - Investigating **indirect gesture input for AR glasses and smart rings**, developing decoders for gesture-center-based and continuous input.
 - Developed **LLM-powered ten-finger typing** on commodity touchscreens through fine-tuning on synthetic and real input data (KeySense, CHI 2026).
 - Built a **VR text-entry system** integrating tap and gesture typing with a cloud-deployed LLM decoder (VRW 2025).
- 07/2023–04/2024 **Research Intern, Simon Fraser University**, Advisor: Xing-Dong Yang
- Developed computational design tools for textile haptics and a battery-free wireless textile sensing interface (CHI 2025).